Conditional Probability & Two-Way Tables

ANSWER KEY

Section 1: Two-Way Frequency Tables

1 A survey of 100 students asks whether they prefer online or in-person learning: 60 are High Schoolers (40 online, 20 in-person) and 40 are Middle Schoolers (15 online, 25 in-person). What fraction of all students surveyed prefer online learning?

a. $\frac{11}{20}$ ✓

b. $\frac{9}{20}$

c. $\frac{2}{5}$

d. $\frac{3}{5}$

Total preferring online: $40 + 15 = 55$. Total students: 100. Fraction: $\frac{55}{100} = \frac{11}{20}$.

2 In the same survey, what is the probability that a student chosen at random is a Middle School student who prefers in-person learning?

a. $\frac{1}{4}$ ✓

b. $\frac{2}{5}$

c. $\frac{3}{10}$

d. $\frac{1}{5}$

Middle school in-person count is 25. Out of 100 total students: $\frac{25}{100} = \frac{1}{4}$.

3 A hospital clinic logs 120 patients: 70 adults (30 tested positive) and 50 children (20 tested positive). What is the probability that a randomly chosen patient tested negative?

a. $\frac{5}{12}$

b. $\frac{7}{12}$ ✓

c. $\frac{1}{2}$

d. $\frac{2}{3}$

Total positive: $30 + 20 = 50$. Total negative: $120 - 50 = 70$. Probability: $\frac{70}{120} = \frac{7}{12}$.

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Section 2: Conditional Probability from Frequency Tables

4 From the student survey (60 High School: 40 online, 20 in-person; 40 Middle School: 15 online, 25 in-person), given that a randomly chosen student prefers online learning, what is the probability that the student is in High School?

a. $\frac{8}{11}$ ✓

b. $\frac{2}{3}$

c. $\frac{3}{11}$

d. $\frac{4}{5}$

Condition is 'prefers online' (total $= 40 + 15 = 55$). Among those, 40 are High School. Probability $= \frac{40}{55} = \frac{8}{11}$.

5 Given that a student is in Middle School, what is the probability that they prefer in-person learning?

a. $\frac{5}{8}$ ✓

b. $\frac{3}{8}$

c. $\frac{1}{4}$

d. $\frac{1}{2}$

Condition is 'Middle School' (total $= 40$). Among Middle Schoolers, 25 prefer in-person. Probability $= \frac{25}{40} = \frac{5}{8}$.

6 At a conference, 80 attendees are engineers (50 use Python) and 40 are managers (10 use Python). Given that an attendee uses Python, what is the probability that they are an engineer?

a. $\frac{5}{6}$ ✓

b. $\frac{2}{3}$

c. $\frac{3}{4}$

d. $\frac{1}{6}$

Total Python users $= 50 + 10 = 60$. Engineers who use Python $= 50$. Probability $= \frac{50}{60} = \frac{5}{6}$.

Section 3: Independent and Dependent Events

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7 Events A and B are independent. If $P(A) = 0.40$ and $P(B) = 0.60$, what is $P(A \text{ and } B)$?

a. 0.24 ✓

b. 0.48

c. 0.52

d. 1.00

For independent events, $P(A \text{ and } B) = P(A) \times P(B) = 0.40 \times 0.60 = 0.24$.

8 A jar contains 6 red marbles and 4 blue marbles. Two marbles are drawn at random without replacement. What is the probability that both marbles are red?

a. $\frac{1}{3}$ ✓

b. $\frac{9}{25}$

c. $\frac{2}{5}$

d. $\frac{1}{2}$

First marble red: $\frac{6}{10}$. Second marble red: $\frac{5}{9}$. Combined probability $= \frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3}$.

9 If $P(A) = 0.5$, $P(B) = 0.7$, and $P(A \text{ and } B) = 0.35$, which of the following is true?

a. A **and** B **are independent events.** ✓

b. A and B are mutually exclusive events.

c. $P(A \mid B) = 0.7$

d. $P(B \mid A) = 0.5$

$P(A) \times P(B) = 0.5 \times 0.7 = 0.35 = P(A \text{ and } B)$. Since the product equals the intersection, A and B are independent.

Section 4: Union & Addition Rule

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10 A standard six-sided die is rolled. What is the probability of rolling an even number OR a number greater than 4?

a. $\frac{1}{2}$

b. $\frac{2}{3}$ ✓

c. $\frac{5}{6}$

d. $\frac{1}{3}$

Even numbers: $\{2, 4, 6\}$ (count 3). Numbers > 4: $\{5, 6\}$ (count 2). Union is $\{2, 4, 5, 6\}$ (count 4). Probability $= \frac{4}{6} = \frac{2}{3}$.

11 If $P(A) = 0.65$, $P(B) = 0.45$, and $P(A \text{ and } B) = 0.25$, what is $P(A \text{ or } B)$?

a. 0.80

b. 0.85 ✓

c. 0.90

d. 1.10

Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.65 + 0.45 - 0.25 = 0.85$.

12 In a group of 50 students, 30 play soccer, 25 play basketball, and 10 play both sports. How many students play neither sport?

a. 5 ✓

b. 10

c. 15

d. 20

Students playing at least one sport: $30 + 25 - 10 = 45$. Students playing neither: $50 - 45 = 5$.

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